

??
 Figure/pteff4e125091112.png Transverse momentum distribution of each of four decayed electrons from 125 GeV Higgs boson
 $p_T < 10$
 p_t
 γ
 γ
 γ
 γ
 η
 ϕ
 E_T
 p_T
 η
 γ
 p_T
 ϕ
 γ
 p_T
 $p_T^e >$
 $|\eta^e| <$
 $dxy <$
 0.5
 $dz <$
 1
 $ZZ^{(*)} \rightarrow$
 4ℓ
 $\Delta R <$
 0.1
 $Z^{(*)}$
 p_T
 p_T
 $5 <$
 $p_T <$
 10
 $p_T >$
 10
 $|\eta|$
 η
 $|\eta| <$
 0.8
 $0.8 <$
 $|\eta| <$
 1.479
 $|\eta| >$
 1.479
 $\eta =$
 0.2
 ??
 Figure/eleIDBDToutput.pdf BDT output for the training and testing sample for true and fake electrons in the two $|\eta| <$
 0.8
 Figure/MVAoutputEB.pdf [width =
 0.45] Figures/chapter3 Figure/MVAoutputEE.pdf Distribution of electron BDT output for training sample $W + jets$, test
 ??
 ??
 ??
 97.6%
 13.2%
 11.5%
 11.5%
 ??
 ??
 Figure/eleIDBDTdata.pdf BDT output distribution on data for electrons with $p_T >$
 10
 ??
 ??
 $\eta\phi\sigma_{i\eta\eta}\sigma_{i\phi i\phi}$
 $\eta\phi$
 H/E
 $(E_{5\times 5} - E_{5\times 1})/E_{5\times 5}$
 R_9
 E_{PS}/E_{raw}
 $E_{tot}/p_{in}E_{ele}/p_{out}1/E - 1/p$
 $\Delta\eta_{in}\Delta\varphi_{in}\Delta\eta_{seed}$
 $f_{brem} = 1 - p_{out}/p_{in}$
 $N_{KF}N_{GSF}(\cdot)$
 $\chi^2\chi_{KF}\chi_{GSF}$
 $(\cdot)$
 $\chi^2(\cdot)$
 $(\cdot)$
 Figure/eleIDROCcurves.pdf ROC curves of the identification classifier for each training category with working point of $|\eta| < 0.80.8 < |\eta| < 1.479|\eta| > 1.479$
 $5 < p_T < 10$
 $p_T > 10$
 Figure/ROCIdOnlyDataEBHighPt.pdf [width =
 0.45] Figures/chapter3 Figure/ROCIdOnlyDataEBHighPt.pdf ROC curves for the electron multivariate identification (EB
 Barrel.Right :
 Endcap
 $|\eta| < 0.80.8 < |\eta| < 1.479|\eta| > 1.479$